Class 14: History of Quantum Mechanics – 1

In the previous class, the possibility of looking at electrons in a solid either as classical particles or as quantum mechanical particles, was considered. Upfront there is no reason to believe that one or the other assumption will be more appropriate. The approach taken is to try out each assumption and compare the predictions made with experimentally obtained data. The approach that results in a better match with experimental data is then validated due to the match.

This brings us to the general topic of Quantum Mechanics. We will now take a step back and examine the origin of quantum mechanics, the history of quantum mechanics, and the major concepts of quantum mechanics. While we can use these concepts directly, the present discussion is a very useful exercise to undertake since it will help put things in perspective. While there is overwhelming proof that nature follows the rules of quantum mechanics, our discussion will enable us to recognize the reasons for our difficulty in coming to terms with it. We will look at the people who opened the door of quantum mechanics for us, and the circumstances under which they led us down this path.

As recently as 1997, a book was published titled "The end of science", by John Horgan, which examined the idea that all the major phenomena that can be discovered in science, have already been discovered. If this were true, many ideas we come across in science fiction, will forever remain in the realm of fiction. Surprisingly, a similar thinking existed almost a hundred years earlier, in the late 19th century. It was believed then that all major discoveries in science had been made, and only minor details remained to be ironed out. The feeling was that there was no real future in physics for any aspiring young person. In hindsight we note that this sense of the state of science at the end of the 19th century, was far from the reality we have seen since. Even today, in the early part of the 21st century, while we know so much more, there is so much that is still to be discovered or understood. For example, scientists who study the nature of the universe and the origins of the universe note that all of the science we know explains only about 20% of what goes to make up our universe. There is an overwhelming 80% of the universe we don't understand yet, which scientists call 'dark energy' and 'dark matter'. So while there are some who understand relativity and quantum mechanics, it is with humility we note that there may be a lot more for us to figure out.

In the late 19th century, when Max Planck went to take up physics, it was generally believed that all major discoveries in Physics had already been made, and only minor details needed to be sorted out. Some of the minor details that remained to be sorted out are listed below:

- Black body radiation
- Discrete nature of atomic and molecular spectra

- Compton modified scattering
- Photoelectric effect

We will spend most of this class focusing on Black Body radiation, let us therefore briefly consider the rest before we proceed.

It was known that when atoms and molecules absorbed energy, they initially gained energy, or were excited, and later released energy to go back to their original state. The energy released by the excited atoms or molecules, appeared only at specific wavelengths and not at all wavelengths. There was no explanation available at that time as to why the energy released was not continuous across all wavelengths but appeared only at specific or discrete wavelengths.

When X-rays interacted with matter, it was observed that after the interaction, the wavelength of the X-ray had increased. This phenomenon, known as Compton modified scattering, could not be explained at the time it was discovered, around the year 1923.

When light was incident on materials, in some cases electrons were ejected. This phenomenon is referred to as the Photoelectric effect. What was observed was that as long as the frequency of the incident radiation was less than a certain threshold value, which varied with materials, no electrons were ejected, no matter how intense the beam of light. At the same time, once the threshold frequency was crossed, electrons were ejected from the material even when the intensity was very low. The impact of the frequency and the lack of impact of the intensity, in initiating the photoelectric effect, was unexplained.

Even though several experimentally observed phenomena such as the above were unexplained, the general belief remained that the knowledge prevalent in the late nineteenth century would only need to be extended marginally and explanations would emerge for these phenomena. There was no expectation that the study of one of these phenomena, the blackbody radiation, would result in the discovery of an entirely new science – quantum mechanics. A discovery that would explain other phenomena as well and fundamentally change our view of the workings of nature.

Let us, therefore, revisit the journey that led to this discovery.

Any body will give out radiation consistent with the temperature it is at. For example, at room temperature, we humans give out infra red (IR) radiation. This is the reason that militaries use IR goggles to spot people at night. At around 1000 ^{o}C, bodies give out visible light, which is how conventional light bulbs function.

When electromagnetic radiation is incident on a body, some of it will be absorbed, some reflected and some transmitted. A body can be imagined and constructed that absorbs all radiation incident on it as long as it cooler than its surroundings. This body will also emit radiation as long as it is hotter than its surroundings. Such a body is referred to as a

'Black body'. Graphite, as a material, comes close to satisfying this description. People tried to design a blackbody, and in 1859 Kirchoff unveiled the design that has since been accepted as a good design for a black body. Figure 14.1 below shows the schematic of the blackbody designed by Kirchoff.

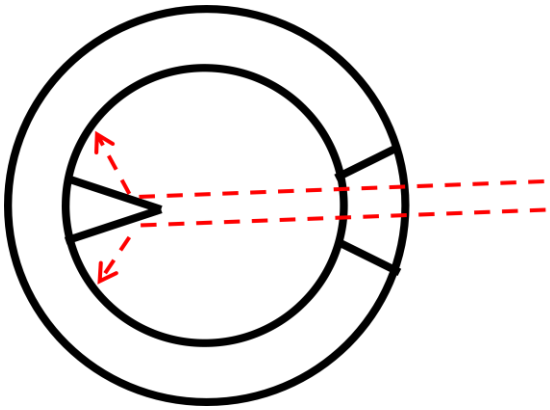


Figure 14.1: Schematic of a blackbody, as designed by Kirchoff. Arrows indicate how radiation entering the body will get absorbed by the internal surfaces of the body.

In general, electromagnetic radiation emitted by a blackbody comes out over a range of wavelengths, however it is not emitted with uniform intensity across all wavelengths. The maximum intensity of the radiation occurs at one wavelength and the intensity decreases for all other wavelengths.

An example of the spectral distribution of the radiation emitted by blackbody is shown in Figure 14.2 below:

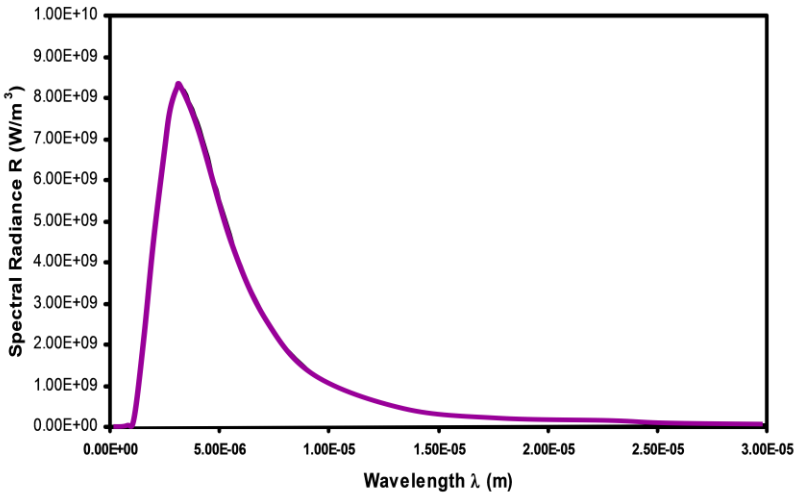


Figure 14.2: The spectral distribution of the radiation emitted by a blackbody

Before we proceed, a short note on the axes in the graph above. The x-axis plots the wavelength in m. The spectral radiance plotted on the y-axis can be understood as follows:

Energy is measured in Joules (J).
Energy per unit area is measured in J/m^2
Power per unit area is represented by $J/m^2/s = W/m^2$
Spectral radiance is power per unit area per unit wavelength and is therefore represented by $W/m^2/m = W/m^3$, which is the unit shown on the y-axis.

Intensity, which is power per unit area, is therefore the area under the curve in Figure 14.2.

Mathematically,

$$I = \int_0^\infty R(\lambda) d\lambda$$

There are two observations that can be made about blackbody radiation:

1) As temperature T of the body increases, intensity of the radiation from the body increases.

2) Higher the temperature, lower is the wavelength of the most intense part of the spectrum.

These two observations are indicated in the schematic in Figure 14.3 below:

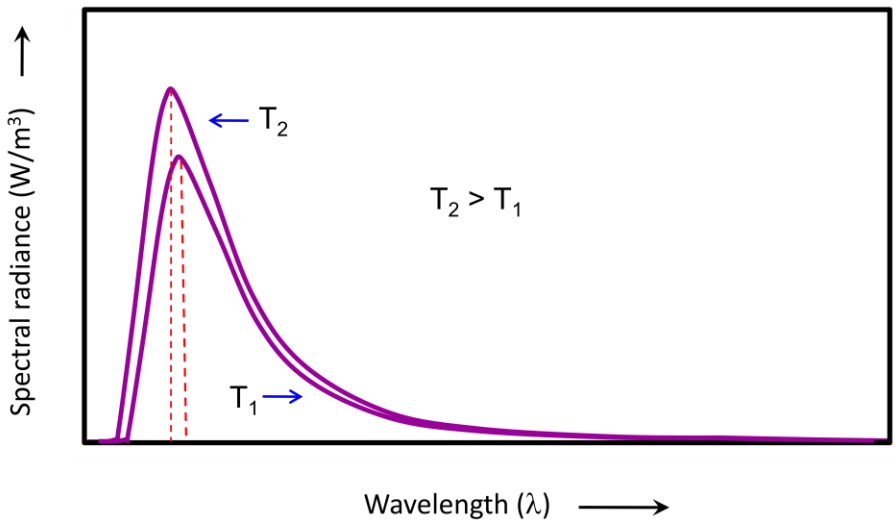


Figure 14.3: Schematic of the variation of blackbody radiation with temperature. At the higher temperature T_2, the area under the curve, and hence intensity, has increased relative to the curve at T_1. At the higher temperature T_2, the wavelength corresponding to the maximum intensity (identified using the red dotted lines in the figure), has decreased relative to that at T_1

These two trends in blackbody radiation, were mathematically stated in the form of two laws:

Stefan Boltzmann Law:

$$I(T) = \sigma T^4$$

where $\sigma = 5.67 \text{ X } 10^{-8} \text{ Wm}^{-2}\text{K}^{-4}$

Wein's displacement Law:

$$\lambda_{max} T = constant = 2898 \ \mu mK$$

The intensity predicted by the Stefan Boltzmann law should match the expression for the area under the curve indicated earlier, therefore:

$$I(T) = \int_0^\infty R(\lambda) d\lambda$$

The scientific challenge that remained was to determine the exact form of the spectral radiance, or power per unit area at a particular wavelength, $R(\lambda)$. Obtaining the equation for $R(\lambda)$, was expected to result in a fundamental understanding of how matter interacted with radiation.

Several researchers worked to determine the form of $R(\lambda)$. One of the early attempts, looked at the matter-radiation interaction in a classical manner, i.e. assumed an equipartition of energy, wherein all modes available to the solid through which it could absorb energy, participated in the process equally. This led to the law known as the Rayleigh -Jean law, which provides an equation for the spectral radiance as follows:

$$R(\lambda, T) = \frac{8\pi kT}{\lambda^4}$$

At higher values of λ, this led to a good match between theory and experiment. However as λ decreases, the theory predicts an ever increasing spectral radiance – a prospect dubbed as 'Ultraviolet catastrophe'. Common experience shows that this does not occur – bodies do not spontaneously release infinite energy. Therefore the Rayleigh-Jean law comprehensively fails at lower wavelengths. However, it was believed that only a minor correction was required to sort out this discrepancy. The mismatch between theory and experiment is shown in the schematic in Figure 14.4 below.

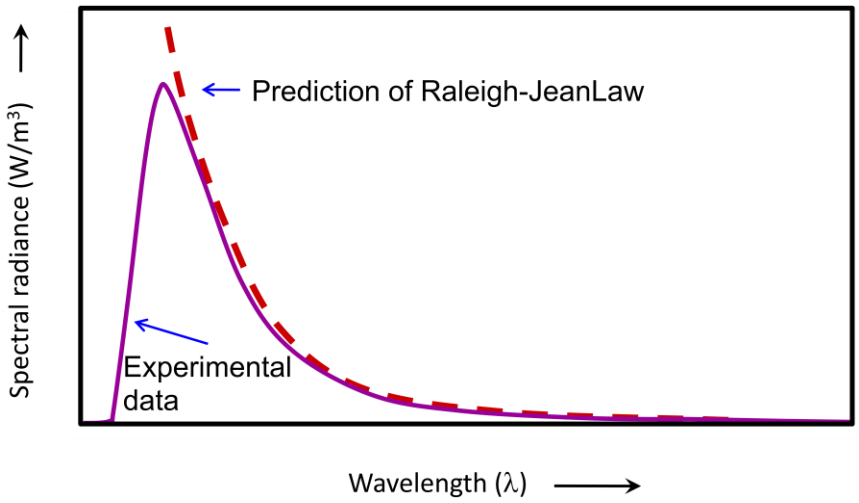


Figure 14.4: a schematic showing the overlay of blackbody radiation data with the prediction of the Rayleigh-Jean law. The theory and data match well at high wavelengths but diverge at lower wavelengths.

Max Planck looked at the data differently and came up with a new expression. He made the assumption that the presence of intensity at any given frequency meant that an oscillator at that frequency was active in the blackbody. He then arbitrarily assumed that for any oscillator to become active in the blackbody, a certain minimum energy was required, although he did not know what that minimum energy was. He placed these assumptions in a mathematical framework saying, with no immediate basis at that time, that the minimum energy required to activate an oscillator of frequency υ was proportional to the frequency υ itself.

In other words, $Energy = Constant \times \upsilon$ was required to activate a single oscillator with the frequency υ. For each additional oscillator at the same frequency υ, additional energy, of the same quantity as above will be required, as per this theory. It is important to note that this was an arbitrary assumption in order to generate a better fit to the experimental data. Planck designated the constant as 'h', and hence $E = h\upsilon$ was required to activate a single oscillator with frequency υ.

This assumption of Planck, although arbitrary, was useful in generating a better curve fit for the experimental data. It created a situation where when the wavelength decreased, and hence frequency increased, the quantity of energy required to activate the oscillator kept increasing with $h\upsilon$. Based on the finite energy available in the system, with decreasing λ, and increasing υ, it would become less likely to impossible to activate the corresponding oscillators, since the $h\upsilon$ would keep increasing to larger values with

increasing v. Therefore the contributions of oscillators to the spectrum would decrease with increasing frequency. Planck's assumption created a situation that enabled higher frequencies to be 'switched off'.

At lower frequencies the step size, or energy increment, hv, was small enough to be switched on with the energy available to the system, a possibility that declined and disappeared at higher frequencies. The energy increments, hv, came to be known as 'quanta'. At that time there was no basis for this type of a model and its assumption of quanta. The model was merely put together to obtain an acceptable curve fit to experimental data.

While his assumptions were arbitrary at that time, Planck enforced the assumption to see what equation he would get for the spectral radiance of a blackbody. Planck did not know the value of the constant h, so while the equation he obtained contained h, he had to vary the value of h till he got a good curve fit to the experimental data. The equation he obtained was as follows:

$$R(\lambda, T) = \frac{8\pi ch}{\lambda^5} \frac{1}{e^{hc/\lambda kT} - 1}$$

The model and its resultant equation fit the experimental data very well. Planck used 'h' as an adjustable parameter to get the model to fit the data and he found that the model matched the predictions with h = 6.55 X 10^{-34} Js. (the value of h that is accepted at present is 6.626 X 10^{-34} Js)

Planck insisted that this was only a model and that there was no reason to believe that the universe actually followed these rules with respect to black body radiation. Inadvertently Planck had stumbled upon the most fundamental rule of what has since evolved as an entirely new field of Physics – Quantum Mechanics.

It is of interest to note that at high λ, we can make the following approximation:

$$e^{hc/\lambda kT} - 1 \approx \left[1 + \frac{hc}{\lambda kT} + \cdots\right] - 1 = \frac{hc}{\lambda kT}$$

which, when substituted into the Planck equation, makes it identical to the Raleigh-Jean equation. Therefore the Planck equation reduces to the Raleigh-Jean equation at higher wavelengths, but at lower wavelengths, or higher frequencies, the Planck equation provides additional detail which the Raleigh-Jean equation does not provide.

While the conventional thinking at that time was that matter and radiation exchanged energy in a continuous manner, Planck suggested that the transaction of energy between matter and radiation had a step size associated with it. It is believed that Planck himself was initially unconvinced that nature behaved in this manner, he had merely made the assumption of quanta to obtain a better fit to the experimental data.

The discovery of Quantum Mechanics is considered profound. Max Planck was awarded the Nobel Prize in 1918 for his discovery. Figure 14.5 below shows a photograph of Max Planck and the citation mentioned for his award.



Max Karl Ernst Ludwig Planck

Nobel Prize in Physics, 1918

"in recognition of the services he rendered to the advancement of Physics by his discovery of energy quanta"

http://nobelprize.org/nobel_prizes/physics/laureates/1918/index.html

Figure 14.5: Photograph of Max Planck and the citation mentioned as part of the Nobel Prize awarded to him.

Blackbody radiation and its analysis is not a matter of esoteric curiosity of a few people. It captures a fundamental piece of information of how matter interacts with energy. Almost a hundred years after Max Planck's work on blackbody radiation, the Nobel Prize in Physics for the year 2006, was awarded to Mather and Smoot for their discovery of the blackbody form of the cosmic microwave background radiation, information that is summarized in Figure 14.6 below.

Based on the discovery of the blackbody form of the cosmic microwave background radiation, it has been possible to estimate that the background temperature of the universe. It is now estimated at 2.75 K. The significance of this estimate is that, we can conclude that this is the lowest temperature that can exist anywhere in the universe, naturally.





John C. Mather **George F. Smoot**

Nobel Prize in Physics, 2006

"for their discovery of the blackbody form and anisotropy of the cosmic microwave background radiation"

http://nobelprize.org/nobel_prizes/physics/laureates/2006/

Figure 14.6: Photographs of John C. Mather and George F. Smoot, and the citation mentioned as part of the Nobel Prize awarded to them.

Using the analysis of blackbody radiation, it has been possible to estimate the temperature of stars that are millions of light years away.

Max Planck's study of blackbody radiation represents the origin of the field of Quantum Mechanics.

In the next class we will look at other important relationships associated with quantum mechanical behavior. We will become familiar with these relationships and see how they relate to each other. Discussion of these relationships is important, because it is this body of relationships that we will take and utilize together when we examine electrons in a solid.